One of the next frontiers of the system is full permissionless-ness. The Foundation, Reown and other collaborators expect to publish a technical paper explaining how the Network will move from its current permissioned state to an open and permissionless state.

4.2 Latency Considerations

Latency in Blockchains and similar permissionless systems is usually constrained as these systems, by design, broadcast all updates to all nodes. Despite innovation in the underlying, often Gossip-based broadcast algorithms and maturity of libraries such as libp2p, making it easier for developers to adopt them, these approaches still cannot overcome the physical constraints of achieving a global Quorum. Distributed Hashtable approaches such as Kademlia improve the average latency, but still suffer from poor worst-case latency. WalletConnect Network's novel permissionless rendezvous hashing works through fulfilling nodes' availability requirements through incentives and keeping reserve and standby nodes.

4.3 Relay Service

The Network's Relay service, used to connect users' wallets to dapps, is by design end-to-end encrypted. The Relay has no insight into users' addresses, transaction hashes, KYC information, or any other information passed between the dapp and wallet. These features make the Network the perfect foundation for future use-cases such as payments where users can trust that middle-men cannot read what they are purchasing on the internet.

4.4 Phases of Decentralization

Initially, the Network operates in a permissioned environment where specific node operators manage the Service Nodes. The Gateway Nodes consume the database formed by the Service Nodes. It is expected that the Gateway Nodes will decentralize in phases similar to the Service Nodes, gradually allowing broader participation and ensuring Network resilience and security.

The Network is designed to operate in a fully permissionless manner, enabling any participant to run a Service Node or Gateway. It is expected that the Network will transition to a permissionless model following community consultation and technical validation to maintain Network integrity and performance.

5 Network Participants

The WalletConnect Network comprises various participants, each playing a critical role in maintaining the Network's functionality and security. This section provides an overview of the key participants and their responsibilities within the Network. The Network protocol comprises Service Nodes and Gateway Nodes.